Worked Example 16 WE16

Write $\log_{10}(10\,000) = 4$ in its equivalent index form.

Thinking	Working
1 Identify the logarithm (power), the base and the numerical value.	4 is the logarithm 10 is the base 10 000 is the numerical value
2 Write the index form.	$10^4 = 10\,000$

12.8 Logarithms

Navigator

Answers page 817

Q1 Column 1, Q2 Column 1, Q3, Q4 Column 1, Q5 Column 1, Q6, Q9, Q11	Q1 Column 2, Q2 Column 2, Q3, Q4 Column 2, Q5 Column 2, Q6, Q7, Q9, Q11	Q1 Column 3, Q2 Column 3, Q3, Q4 Column 3, Q5 Column 3, Q6, Q8, Q9, Q10

Equipment required: calculator for Questions **4–5**

Fluency

WE15

1 Write each of the following in its equivalent logarithmic form.

(a) $2^5 = 32$ **(b)** $2^8 = 256$ **(c)** $2^{12} = 4096$

(d) $3^3 = 27$ **(e)** $3^5 = 243$ **(f)** $3^8 = 6561$

(g) $10^2 = 100$ **(h)** $10^6 = 1\,000\,000$ **(i)** $10^8 = 100\,000\,000$

(j) $2^{-2} = \frac{1}{4}$ **(k)** $2^{-3} = \frac{1}{8}$ **(l)** $2^{-4} = \frac{1}{16}$

WE16

2 Write each of the following in its equivalent index form.

(a) $\log_{10}100 = 2$ **(b)** $\log_{10}1000 = 3$ **(c)** $\log_{10}100\,000 = 5$

(d) $\log_2 16 = 4$ **(e)** $\log_2 128 = 7$ **(f)** $\log_2 512 = 9$

(g) $\log_{10}\left(\frac{1}{1000}\right) = -3$ **(h)** $\log_{10}\left(\frac{1}{10\,000}\right) = -4$ **(i)** $\log_{10}\left(\frac{1}{1\,000\,000}\right) = -6$

(j) $\log_2\left(\frac{1}{8}\right) = -3$ **(k)** $\log_2\left(\frac{1}{32}\right) = -5$ **(l)** $\log_2\left(\frac{1}{128}\right) = -7$

3 **(a)** If $\log_{10}x = -2$ then x is equal to:

A -200 **B** $-\frac{1}{100}$ **C** $\frac{1}{100}$ **D** 200

(b) Writing $2^x = 2048$ in logarithmic form gives:

A $\log_2 x = 2048$ **B** $\log_x 2 = 2048$ **C** $\log_x 2048 = 2$ **D** $\log_2 2048 = x$

Understanding

4 Solve the following for x:

(a) $3^x = 81$ (b) $3^x = 6561$ (c) $3^x = 19\,683$

(d) $5^x = \frac{1}{25}$ (e) $5^x = \frac{1}{125}$ (f) $5^x = \frac{1}{15\,625}$

(g) $6^{-x} = 216$ (h) $6^{-x} = 7776$ (i) $6^{-x} = \frac{1}{1296}$

> **Hint**
> Write the number on the RHS as an index number, using the same base as the LHS.

5 Solve the following for x:

(a) $\log_2 x = 4$ (b) $\log_2 x = 6$ (c) $\log_2 x = 7$

(d) $\log_3\left(\frac{1}{81}\right) = x$ (e) $\log_3\left(\frac{1}{243}\right) = x$ (f) $\log_3\left(\frac{1}{6561}\right) = x$

(g) $\log_x 625 = 4$ (h) $\log_x 343 = 3$ (i) $\log_x\left(\frac{1}{3125}\right) = -5$

Reasoning

6 (a) Solve for x in each of the following:

(i) $\log_{10} x = 1$ (ii) $\log_5 x = 1$ (iii) $\log_7 x = 1$

(b) Write a general rule based on the above, using a as your base.

7 (a) Solve for x in each of the following:

(i) $\log_{10} x = 0$ (ii) $\log_5 x = 0$ (iii) $\log_7 x = 0$

(b) Write a general rule to explain what you have just found out.

8 In the equation $\sqrt{a} = a^{\frac{1}{2}}$, why does a have to be greater than or equal to 0?

9 (a) Check that both of the following logarithmic equations are correct:

$\log_2 64 = 6$ and $\log_4 64 = 3$

(b) Now, given $\log_2(4096) = 12$, find the value of y in $\log_4(4096) = y$.

(c) Explain how you got your answer for **(b)**.

Open-ended

10 Given $\log_3(531\,441) = 12$, write two other related log statements that are true.

11 Lee wrote:

$-(6)^2 = -36$

because $-\log_6(-36) = 2$

Explain why this is incorrect and how the question should be tackled.

1 **(a)** Complete the Richter scale chart below to show the comparative strength of earthquakes.

Richter scale	0	1	2	3	4	5	6	7	8	9
Comparative strength	1	10^1	10^2							

(b) (i) How many times stronger is an earthquake with magnitude 8.0 compared to an earthquake with magnitude 5.0?

(ii) How many times stronger is an earthquake with magnitude 6.5 compared to an earthquake with magnitude 7.5?

(c) Draw a graph showing the comparative strength of earthquakes on the Richter scale.

(d) What type of graph is this?

(e) Write an equation to describe this graph.

(f) Use the equation to find the comparative strength of an earthquake with magnitude 5.2. What does this figure mean?

2 The Earth has seven major tectonic plates that can cause earthquakes when sliding past each other. New Zealand, Indonesia, Japan, China, the west coast of the USA and South America all lie directly above where two tectonic plates meet and therefore are more susceptible to frequent and severe earthquakes.

(a) Show the position of the following deadly earthquakes on the graph drawn in **1(c)**.

- Sichuan earthquake, China, 2008, magnitude of 7.9, killing 88 000 people
- The Great Kanjō Earthquake, Japan, 1923, magnitude of 8.3, killing 140 000 people
- The Great Hanshin Earthquake, Kobe, Japan, 1995, magnitude of 7.2, killing 6433 people
- Java earthquake, Indonesia, May 2006, magnitude of 6.4, killing 5749 people
- Christchurch, New Zealand, February 2011, magnitude of 6.3, killing 181 people

One day after the Sichuan earthquake in China, another earthquake hit China with a magnitude of 6.1. The difference in strength between the two earthquakes can be found by calculating the ratio of the strengths of the two earthquakes using the following method:

$$\frac{\text{Magnitude}_1}{\text{Magnitude}_2} = \frac{10^{7.9}}{10^{6.1}} = 10^{1.8} = 63$$

The Sichuan earthquake was 63 times stronger than the earthquake the following day.

(b) Calculate the differences in strength between the following earthquakes listed in **3(a)**.

(i) The Great Kanjō Earthquake and The Great Hanshin Earthquake

(ii) The 2006 Java earthquakes and the 2011 Christchurch earthquake.

3 In 1906 an earthquake hit San Francisco, leaving 3000+ people dead and starting fires that almost flattened the whole city. This earthquake was 8 times stronger than the 1989 San Francisco earthquake that measured 6.9 on the Richter scale. The information above can be shown as:

$$\frac{\text{Magnitude}_1}{\text{Magnitude}_2} = \frac{10^y}{10^{6.9}} = 10^x = 8$$

$10^x = 8$ can be written as $\log_{10} 8 = x$

This can be solved on a calculator:

$\log_{10} 8 = 0.9$ (1 d.p.)

$$\frac{\text{Magnitude}_1}{\text{Magnitude}_2} = \frac{10^y}{10^{6.9}} = 10^{0.9} = 8$$

Therefore, the magnitude of the 1906 San Francisco earthquake had a magnitude of 7.8. Between 12 September 2007 and 13 September 2007, three earthquakes hit Indonesia, with the first registering a magnitude of 8.5. This was 4 times stronger than the second earthquake 12 hours later.

(a) What was the magnitude of the second earthquake?

(b) The first earthquake was 25 times stronger than the third earthquake. What was the magnitude of the third earthquake?

4 An earthquake in Taiwan in 2006 had a magnitude of 7.0, the same as recorded at Haiti in the 2010 earthquake. Over 230 000 people were killed in Haiti, whereas only two were killed in the Taiwan earthquake. Give reasons for the differences in devastation caused by the two similar strength earthquakes.

Research

5 The following measures also involve exponentials:

- loudness of sound (decibels)
- brightness of light
- acidity levels (pH readings)

Investigate how these scales are exponential and how they produce accurate and comparative readings.